

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, adherent	Molecules Electroporated	DNA: supercoiled plasmid; luciferase reporter gene
Species Used	Rat, CA77, medullary thyroid carcinoma cell line		

Before the Pulse

Cell Growth Medium	DMEM/F12 (1:1) + 10 % Fetal Bovine Serum (GIBCO/BRL, Sigma)	Growth Phase at Harvest	Active growth, about 50 to 70 % confluent
Wash Solution	Dulbecco's Phosphate Buffered Saline (minus Ca++, Mg++)	Pre-pulse Incubation	None

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature	Room temperature	Cuvette Gap	0.4 cm
Electroporation Medium*	Dulbecco's Phosphate Buffered Saline (minus Ca++, Mg++)	Voltage	0.22 kV
Cell Density	4 to 5 x10 (6) cells / 800 µl	Field Strength	0.55 kV/cm
Volume of Cells	800 µl	Capacitor	960 µF
DNA Concentration	5 to 20 µg	Resistor	(Pulse Controller) Ω none
DNA Resuspension Buffer	TE (10 mM Tris, 1 mM EDTA)	Time Constant	about 12 msec
Volume of DNA	5 to 50 µl (usually less than 10 µl)		
After the Pulse			
Outgrowth Medium	DMEM/F12 (1:1) + 10 % Fetal Bovine Serum		

Outgrowth Temperature	37 °C	Relevant Publications and/or Comments Note: exponential values designated in parentheses. Reference: Tverberg, L.A. and Russo, A.F. 1992. <i>J. Biol. Chem.</i> 267 (5):17567-17573.
Length of Incubation	usually 24 hours	
Selection Method or Assay Used	Luciferase assay, β-galactosidase assay	
Electroporation Efficiency	about 1 x 10 (4) / µg [20 to 40 % cells stained blue, b-gal, transient assay]	
Per Cent Survival	50 to 80 %	

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